

Aditi Bais

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Summary

M.Tech (Information Technology) student with hands-on experience building Generative AI applications using Python, LangChain, LLMs, Retrieval-Augmented Generation (RAG), and Computer Vision. Developed AI chatbots, AI agents, and deep learning projects using Streamlit, PyTorch, OpenCV, and modern LLM frameworks. Seeking an AI Internship to contribute to real-world intelligent automation and AI solutions.

Technical Skills

Languages: Python, JavaScript, SQL.

AI/ML: LangChain, PyTorch, TensorFlow/Keras, Scikit-learn, OpenCV, Hugging Face, Groq, RAG, FAISS.

Frameworks: Flask, Streamlit.

Databases: PostgreSQL, Oracle, ChromaDB.

Tools: Git, GitHub, Jupyter Notebook, PyCharm.

Operating Systems: Linux, Windows.

Projects

- 🌐 **KisanSaathi** | *Python, LangChain, Groq, ChromaDB, PyTorch, Streamlit* Jul 2026 – Ongoing
 - Designing a multilingual (Hindi/English) agricultural chatbot using a RAG pipeline over a curated crop knowledge base to deliver sowing, irrigation, and crop-selection guidance.
 - Building a CNN-based crop disease detection module to identify deficiencies and infections from leaf images(Ongoing).
- 🌐 **Emotion-Based Music Recommender** | *Python, TensorFlow/Keras, OpenCV, Spotipy* Sept 2025 – Nov 2025
 - Built a CNN-based facial emotion detection model to classify user mood in real-time using webcam input via OpenCV.
 - Integrated the Spotipy API to generate mood-matched song recommendations, mapping detected emotions to curated Spotify playlists.
- 🌐 **YouTube RAG Chatbot** | *Python, LangChain, YouTube Transcript API, FAISS, Streamlit* June 2026 – July 2026
 - Built a RAG pipeline using LangChain to enable question-answering over YouTube video content, covering document loading, chunking, embedding generation, similarity-based retrieval, and response generation.
 - Fetched video transcripts via the YouTube Transcript API and developed a Streamlit interface allowing users to paste a video link and chat with its content in real time.
- 🌐 **Mental Health Companion Chatbot** | *Python, LangChain, Groq API, Hugging Face Transformers, Streamlit* June 2026
 - Built an empathetic conversational chatbot using a decoder-only LLM (Groq-hosted) combined with a fine-tuned model for real-time emotion classification from user input.
 - Engineered dynamic prompts injecting detected emotional context and safety guidelines to generate emotionally-aware, non-diagnostic responses, with conversation memory maintained across turns via Streamlit session state.
- 🌐 **Multi-Tool AI Agent** | *Python, LangChain, Tool Calling* Jul 2026
 - Built an autonomous LangChain agent that dynamically selects between a calculator tool and a live weather API based on user query intent, using function/tool calling to ground responses in accurate, real-time data.
 - Designed a tool-calling architecture to offload precision-dependent tasks from the LLM, addressing hallucination risks inherent in probabilistic text generation for arithmetic and time-sensitive information.

Education

International Institute of Professional Studies	2022 – Present
<i>M.Tech in Information Tecnology - CGPA: 8.87/10</i>	<i>Indore 2022</i>
Sideras International School	
<i>Senior Secondary Certificate (CBSE Class 12th) - Percentage: 89.4%</i>	<i>Indore 2020</i>
Sideras International School	
<i>Secondary School Certificate (CBSE Class 10th) - Percentage: 89.6%</i>	<i>Indore</i>

Certifications

Data Science Training <i>Certificate - Acmegrade & Mood Indigo, IIT Bombay</i>	Dec 2023
IoT Based Product Development <i>Certificate - IIPS DAVV & UPSYS IT & Marketing Solutions</i>	Nov 2024